

Generalized Linear Models

Introduction

We developed and illustrated data transformation as an approach to fitting regression models when the assumptions of a normally distributed response variable with constant variance are not appropriate.

Transformation of the response variable is often a very effective way to deal with both response nonnormality and inequality of variance.

Weighted least squares is also a potentially useful way to handle the non-constant variance problem.

In this lesson, we present an alternative approach to data transformation when the “usual” assumptions of normality and constant variance are not satisfied.

This approach is based on the generalized linear model (GLM).

The GLM is a unification of both linear and nonlinear regression models that also allows the incorporation of nonnormal response distributions.

In a GLM, the response variable distribution must only be a member of the exponential family, which includes the normal, Poisson, binomial, exponential, and gamma distributions as members.

Logistic Regression Models

Consider the situation where the response variable in a regression problem takes on only two possible values, 0 and 1. These could be arbitrary assignments resulting from observing a qualitative response.

Suppose that the model has the form:

$$y_i = \mathbf{x}_i' + \varepsilon_i$$

where $\mathbf{x}_i' = [1, x_{i1}, x_{i2}, \dots, x_{ik}]$, $\beta = [\beta_0, \beta_1, \dots, \beta_k]$ and the response variable y_i takes on the value either 0 or 1.

We will assume that the response y_i is a **Bernoulli Random Variable** with the probability distribution as follows:

y_i	Probability
1	$P(y_i = 1) = \pi_i$
0	$P(y_i = 0) = 1 - \pi_i$

Now since $E(\varepsilon_i) = 0$ the expected value of the response variable is:

$$E(y_i) = 1(\pi_i) + 0(1 - \pi_i) = \pi_i$$

This implies that:

$$E(y_i) = \mathbf{x}_i\beta = \pi_i$$

This means that the expected response given by the response function $E(y_i) = \mathbf{x}_i' \beta$ is just the probability that the response variable takes on the value 1.

Generally, when the response variable is binary, there is considerable empirical evidence indicating that the shape of the response function should be nonlinear.

This function is called the logistic response function and has the form:

$$E(y) = \frac{\exp(\mathbf{x}'\beta)}{1 + \exp(\mathbf{x}'\beta)} = \frac{1}{1 + \exp(-\mathbf{x}'\beta)}$$

The logistic response function can be easily linearized.

One approach defines the structural portion of the model in terms of a function of the response function mean.

Let

$$\eta = \mathbf{x}'\beta$$

be the **linear predictor** where η is defined by the transformation

$$\eta = \ln \frac{\pi}{1 - \pi}$$

This transformation is often called **logit transformation** of the probability π and the ratio $\frac{\pi}{1-\pi}$ in the transformation is called the odds.

Estimating the Parameters in a Logistic Regression Model

The general form of the logistic regression model is:

$$y_i = E(y_i) + \varepsilon_i \quad (27)$$

where the observations y_i are independent Bernoulli random variables with expected values

$$E(y_i) = \pi = \frac{\exp(\mathbf{x}'\beta)}{1 + \exp(\mathbf{x}'\beta)}$$

We use the method of maximum likelihood to estimate the parameters in the linear predictor $\mathbf{x}_i' \beta$

Each sample observation follows the Bernoulli distribution so the probability distribution of each sample observation is:

$$f_i y_i = \pi_i^{y_i} (1 - \pi_i)^{(1-y_i)}, i = 1, 2, \dots, n$$

and of course each observation y_i takes on the values 0 or 1.

Since the observation are independent the likelihood function is just:

$$L(y_1, y_2, \dots, y_n, \beta) = \prod_{i=1}^n f_i y_i = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \quad (28)$$

It is more convenient to work with the log-likelihood

$$\begin{aligned}\ln L(y_1, y_2, \dots, y_n, \beta) &= \ln \prod_{i=1}^n f_i y_i \\ &= \sum_{i=1}^n \left[y_i \ln \left(\frac{\pi_i}{1 - \pi_i} \right) \right] + \sum_{i=1}^n \ln(1 - \pi_i)\end{aligned}\quad (29)$$

Now since $1 - \pi_i = [1 + \exp(x'_i \beta)]^{-1}$ and $\eta_i = \ln \left[\frac{\pi_i}{(1 - \pi_i)} \right] = x'_i \beta$ the log-likelihood can be written as

$$\ln L(y, \beta) = \sum_{i=1}^n y_i x'_i \beta - \sum_{i=1}^n \ln[1 + \exp(x'_i \beta)] \quad (30)$$

Often in logistic regression models we have repeated observations or trials at each level of the x variables.

Let y_i represent the number of 1's observed for the i^{th} and n_i be the number of trials at each observation.

Then the log-likelihood becomes:

$$\begin{aligned}\ln L(y, \beta) &= \sum_{i=1}^n y_i \ln(\pi_i) + \sum_{i=1}^n n_i \ln(1 - \pi_i) - \sum_{i=1}^n y_i \ln(1 - \pi_i) \\ &= \sum_{i=1}^n y_i \ln(\pi_i) + \sum_{i=1}^n (n_i - y_i) \ln(1 - \pi_i)\end{aligned}\quad (31)$$

Numerical search methods could be used to compute the maximum-likelihood estimates (or MLEs) $\hat{\beta}$

The estimated value of the linear predictor is $\hat{\eta}_i = x'_i \hat{\beta}$ and the fitted values of the logistic regression model is written as:

$$\hat{y}_i = \hat{\pi}_i = \frac{\exp(\hat{\eta}_i)}{1 + \exp(\hat{\eta}_i)} = \frac{\exp(x'_i \hat{\beta})}{1 + \exp(x'_i \hat{\beta})} = \frac{1}{1 + \exp(-x'_i \hat{\beta})}$$

Interpretation of Parameters in a Logistic Regression Model

Consider the first case where the linear predictor has only a single predictor so that the fitted values of the linear predictor at a particular value of x say x_i is:

$$\hat{\eta}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

The fitted value at $x_i + 1$ is

$$\hat{\eta}(x_i + 1) = \hat{\beta}_0 + \hat{\beta}_1 (x_i + 1)$$

and the difference between the two predicted values is:

$$\hat{\eta}(x_i + 1) - \hat{\eta}(x_i) = \hat{\beta}_1$$

Now $\hat{\eta}(x_i)$ is just the log-odds when the regressor variable is equal to x_i and $\hat{\eta}(x_i + 1)$ is just the log-odds when the regressor is equal to $x_i + 1$.

Therefore the difference in the two fitted values is:

$$\hat{\eta}(x_i + 1) - \hat{\eta}(x_i) = \ln(odds_{x_i+1}) - \ln(odds_{x_i}) = \ln\left(\frac{odds_{x_i+1}}{odds_{x_i}}\right) = \hat{\beta}_1$$

The odds ratio can be interpreted as the estimated increase in the probability of success associated with a one-unit change in the value of the predictor variable.

Example of Logistic Modeling

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable.

The description of the data is given as:

```
mydata <- read.csv("https://stats.idre.ucla.edu/stat/data/binary.csv")
knitr::kable(head(mydata))
```

The dataset has a binary response variable called **admit**

Variables **gre** and **gpa** are continuous

Variable **rank** takes on the values 1 through 4.

We can fit the model as:

```
# convert the rank to factor

mydata$rank <- factor(mydata$rank)

# fitting the model

model <- glm(admit ~ gre + gpa + rank, data = mydata, family = "binomial")

# output
library(dplyr)
model %>%
  broom::tidy()
```

The logistic regression coefficients give the change in the log odds of the outcome for a one unit increase in the predictor variable.

- For every one unit change in gre, the log odds of admission (versus non-admission) increases by 0.002
- For a one unit increase in gpa, the log odds of being admitted to graduate school increases by 0.804.

- The indicator variables for rank have a slightly different interpretation. For example, having attended an undergraduate institution with rank of 2, versus an institution with a rank of 1, changes the log odds of admission by -0.675.

You can also exponentiate the coefficients and interpret them as odds-ratios. Now we can say that for a one unit increase in gpa, the odds of being admitted to graduate school (versus not being admitted) increase by a factor of 2.23.

```
library(dplyr)
model %>%
  broom::tidy(exp = TRUE)
```

We can look at the overall quality of the model with the glance() function

```
library(dplyr)
model %>%
  broom::glance()
```

Poisson Regression

We assume that the response variable y_i is a count, such that the observation $Y_i = 0, 1, 2, \dots$. A reasonable probability model for count data is often the Poisson distribution.

$$f(y) = \frac{e^{-\mu} \mu^y}{y!}, y = 0, 1, \dots$$

where the parameter $\mu > 0$.

For the Poisson distribution

$$E(Y) = \mu = Var(Y)$$

The Poisson regression model can be written as:

$$Y_i = E(Y_i) + \varepsilon_i, i = 1, 2, \dots, n$$

We assume that the expected value of the observed response can be written as:

$$E(y_i) = \mu_i$$

and there is a function g that relates the mean response to a linear predictor say

$$g(\mu_i) = \eta_i = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k = \mathbf{x}_i' \boldsymbol{\beta}$$

The function g is usually called the **link function**.

The relationship between the mean and the linear predictor is

$$\mu_i = g^{-1}(\eta_i) = g^{-1}(x_i' \boldsymbol{\beta})$$

The link function for the Poisson distribution is the **log link**